

Geethu's Corner — Learning Roadmap

Concepts to study before and during the rebuild. Stack: React.js · Spring Boot · PostgreSQL · FastAPI.
Tick each box as you finish it.

LEVEL 1

Must know

Learn these first. The project cannot be built without them.

LEVEL 2

Should know

Learn these while building. Needed for a real, job-ready project.

LEVEL 3

Basic awareness

Know what they are. Learn deeply later or when needed.

Suggested study order

- 1 Web Fundamentals (Section 1)
- 2 Spring Boot Core refresh: DTOs, validation, exception handling (Section 4)
- 3 JPA relationships (Section 5)
- 4 Spring Security + JWT (Section 6)
- 5 Email & OTP (Section 7)
- 6 React Router, Context API, Axios (Section 3)
- 7 Tailwind CSS (Section 3)
- 8 Git & Deployment (Section 11)
- 9 AI Service (Section 10)

1. Web Fundamentals

Needed for every part of the project

LEVEL 1

Must know

- ☐ HTTP methods: GET, POST, PUT, PATCH, DELETE
- ☐ HTTP status codes: 200, 201, 400, 401, 403, 404, 500
- ☐ REST API design (resource URLs, nouns not verbs)
- ☐ JSON format
- ☐ Request / response headers
- ☐ Frontend ↔ backend communication flow

LEVEL 2

Should know

- ☐ CORS (why the browser blocks requests)
- ☐ localStorage vs cookies vs sessionStorage
- ☐ Authorization header (Bearer token)
- ☐ Browser DevTools: Network & Console tabs

LEVEL 3

Basic awareness

- ☐ HTTPS & SSL basics
- ☐ Domain & DNS basics
- ☐ Client-side vs server-side validation

2. JavaScript (ES6+)

The base for React

LEVEL 1

Must know

- ☐ let / const, arrow functions
- ☐ Destructuring (objects & arrays)
- ☐ Spread & rest operator
- ☐ Array methods: map, filter, reduce, find
- ☐ Template literals
- ☐ ES modules: import / export
- ☐ Promises, async / await, try / catch
- ☐ fetch API

LEVEL 2

Should know

- ☐ Optional chaining (? .) & nullish (??)
- ☐ Date handling
- ☐ FormData object
- ☐ setTimeout / setInterval (OTP resend timer)

LEVEL 3

Basic awareness

- ☐ Debouncing (search box)
- ☐ Event loop basics

3. React (Frontend)

React.js + Vite + Tailwind CSS

LEVEL 1

Must know

- ☐ Components & props
- ☐ useState
- ☐ useEffect (with dependency array)
- ☐ Conditional rendering
- ☐ Lists & keys
- ☐ Controlled form inputs
- ☐ Event handling
- ☐ React Router: Routes, Route, Link, useNavigate, useParams
- ☐ Protected routes (login / admin only)
- ☐ Context API + useContext (auth & cart)
- ☐ Axios: GET / POST / PUT / DELETE
- ☐ Loading & error states
- ☐ Persisting cart / token in localStorage

LEVEL 2

Should know

- ☐ Custom hooks
- ☐ Axios instance & interceptors (attach JWT)
- ☐ Form validation (React Hook Form + Zod)
- ☐ Tailwind: utility classes, flex, grid, responsive (sm/md/lg)
- ☐ Vite env variables (import.meta.env)
- ☐ useRef
- ☐ Lifting state up
- ☐ File upload with FormData + image preview
- ☐ Layout components (Navbar, Footer, AdminLayout)

LEVEL 3
Basic awareness

- ☐ TanStack Query (React Query)
- ☐ Toast notifications (react-hot-toast)
- ☐ Lazy loading & code splitting
- ☐ useMemo / useCallback
- ☐ Accessibility basics (alt, labels)
- ☐ SEO basics (title, meta tags)
- ☐ ESLint & Prettier

4. Java & Spring Boot Core

The backend foundation

LEVEL 1
Must know

- ☐ Java OOP: class, interface, inheritance
- ☐ Collections: List, Map, Optional
- ☐ Exceptions & custom exceptions
- ☐ Spring Boot project structure
- ☐ Layered architecture: Controller → Service → Repository
- ☐ Dependency Injection (constructor injection)
- ☐ @RestController, @RequestMapping
- ☐ @GetMapping / @PostMapping / @PutMapping / @DeleteMapping
- ☐ @PathVariable, @RequestParam, @RequestBody
- ☐ ResponseEntity & HTTP status
- ☐ DTOs (request / response) vs Entities
- ☐ application.properties & environment variables
- ☐ Lombok annotations

LEVEL 2
Should know

- ☐ Bean Validation: @Valid, @NotBlank, @Email, @Size, @Pattern
- ☐ Global exception handling: @RestControllerAdvice
- ☐ Spring profiles (dev / prod)
- ☐ Pagination & sorting (Pageable)
- ☐ DTO ↔ Entity mapping
- ☐ @Transactional
- ☐ Enums
- ☐ LocalDate / LocalDateTime
- ☐ Java Streams basics
- ☐ Logging with SLF4J
- ☐ Java records

LEVEL 3
Basic awareness

- ☐ Swagger / OpenAPI (springdoc)
- ☐ Spring Boot Actuator (health check)
- ☐ @Scheduled tasks
- ☐ @Async
- ☐ RestClient / WebClient (calling the AI service)
- ☐ Maven: pom.xml, mvn clean package

5. Database: JPA & PostgreSQL

Storing users, cakes, orders, reviews

LEVEL 1
Must know

- ☐ SQL: SELECT, INSERT, UPDATE, DELETE
- ☐ WHERE, ORDER BY, GROUP BY, JOIN
- ☐ Primary key & foreign key
- ☐ @Entity, @Table, @Id, @GeneratedValue, @Column
- ☐ JpaRepository built-in methods
- ☐ Derived query methods (findByEmail ...)
- ☐ @ManyToOne & @OneToMany
- ☐ Connecting Spring Boot to PostgreSQL

LEVEL 2
Should know

- ☐ Fetch types: LAZY vs EAGER
- ☐ Cascade & orphanRemoval
- ☐ @Query (JPQL)
- ☐ Unique constraints
- ☐ ddl-auto: update vs validate
- ☐ @CreationTimestamp / @UpdateTimestamp
- ☐ BigDecimal for money
- ☐ The N+1 query problem

LEVEL 3
Basic awareness

- ☐ Flyway database migrations
- ☐ Indexes
- ☐ Normalization basics
- ☐ Connection pooling (HikariCP)
- ☐ Supabase dashboard

6. Spring Security & Authentication

The most important new topic for you

LEVEL 1

Must know

- ☐ Authentication vs Authorization
- ☐ Password hashing: BCrypt, PasswordEncoder
- ☐ SecurityFilterChain configuration
- ☐ Stateless sessions
- ☐ JWT: header, payload, signature, expiry
- ☐ Generating & validating JWT (jjwt library)
- ☐ JWT filter (OncePerRequestFilter)
- ☐ UserDetails & UserDetailsService
- ☐ Roles: CUSTOMER & ADMIN
- ☐ requestMatchers, permitAll, hasRole
- ☐ Signup & login flow
- ☐ CORS configuration in Spring Security

LEVEL 2

Should know

- ☐ @PreAuthorize (method security)
- ☐ AuthenticationManager
- ☐ SecurityContextHolder (current logged-in user)
- ☐ 401 vs 403 error handling
- ☐ Refresh tokens
- ☐ Token storage: localStorage vs httpOnly cookie
- ☐ Logout
- ☐ Rate limiting login / OTP attempts

LEVEL 3

Basic awareness

- ☐ CSRF (and why JWT APIs disable it)
- ☐ OAuth2 / Google login
- ☐ Account logout
- ☐ OWASP Top 10 basics

7. Email & OTP Verification

Signup verification + forgot password

LEVEL 1

Must know

- ☐ OTP flow: signup → generate → store → email → verify → enable account
- ☐ SecureRandom (generating the code)
- ☐ OTP expiry with LocalDateTime
- ☐ User 'verified' flag
- ☐ Sending email through an HTTP API (Brevo)
- ☐ Forgot password / reset password flow

LEVEL 2

Should know

- ☐ Resend OTP with cooldown
- ☐ Maximum wrong attempts
- ☐ Hashing the stored OTP
- ☐ HTML email templates
- ☐ Order status update emails

LEVEL 3

Basic awareness

- ☐ SMTP vs email API
- ☐ Thymeleaf for email templates
- ☐ Email deliverability (spam folder)
- ☐ SPF / DKIM records

8. File Uploads & Images

Cake photos & custom cake reference images

LEVEL 1

Must know

- ☐ MultipartFile in Spring Boot
- ☐ Sending files from React with FormData
- ☐ Uploading to Cloudinary & saving the image URL

LEVEL 2

Should know

- ☐ File size & type validation
- ☐ Cloudinary image transformations (resize, compress)

LEVEL 3

Basic awareness

- ☐ Signed uploads
- ☐ Deleting old images

9. Project Business Logic

How a cake shop actually works in code

LEVEL 1

Must know

- ☐ Cart → Order conversion
- ☐ Calculating the total on the server (never trust the client)
- ☐ Order status lifecycle using an enum
- ☐ Delivery vs pickup
- ☐ Date slots & daily order limit

LEVEL 2

Should know

- ☐ Custom cake: request → quote → accept flow
- ☐ Reviews allowed only after delivery
- ☐ Admin dashboard stats (COUNT / SUM queries)

LEVEL 3

Basic awareness

- ☐ Two customers booking the last slot at once (concurrency)
- ☐ Soft delete (hide instead of delete)

10. AI Service (Python + FastAPI)

The cake shop chatbot

LEVEL 1

Must know

- ☐ Python basics: functions, lists, dicts
- ☐ Virtual environment, pip, requirements.txt
- ☐ FastAPI routes & Pydantic models
- ☐ Running with uvicorn
- ☐ .env & python-dotenv
- ☐ Calling Gemini through LangChain
- ☐ RAG concept: load → split → embed → store → retrieve

LEVEL 2

Should know

- ☐ Fetching the live menu from the Spring Boot API (httpx / requests)
- ☐ Prompt engineering: system prompt & guardrails
- ☐ Chat history / memory
- ☐ async vs sync endpoints
- ☐ CORS in FastAPI

LEVEL 3

Basic awareness

- ☐ LangGraph agents & tools
- ☐ Vector databases (Chroma / pgvector)
- ☐ Streaming responses
- ☐ Token limits & API quotas

11. Git, Deployment & DevOps

Making it live on the internet

LEVEL 1

Must know

- ☐ Git: init, add, commit, push, pull
- ☐ Branches
- ☐ .gitignore
- ☐ GitHub repository
- ☐ Environment variables & secrets (never commit them)
- ☐ Deploying React on Vercel
- ☐ Deploying Spring Boot on Render
- ☐ Creating the PostgreSQL database on Supabase

LEVEL 2

Should know

- ☐ Docker basics: Dockerfile, build, run
- ☐ Multi-stage Docker build
- ☐ npm run build / mvn package
- ☐ Reading logs on Render
- ☐ Vercel SPA rewrites (React Router refresh fix)

LEVEL 3

Basic awareness

- ☐ GitHub Actions (CI)
- ☐ docker-compose for local development
- ☐ Uptime monitoring (UptimeRobot)
- ☐ Free tier limits (sleeping, hours)
- ☐ Custom domain setup

12. Testing

Proving your code works (interviewers ask this)

LEVEL 1

Must know

- ☐ Testing APIs with Postman / Thunder Client
- ☐ Debugging with Browser DevTools

LEVEL 2

Should know

- ☐ JUnit 5
- ☐ Mockito (mocking repositories)
- ☐ @WebMvcTest / @SpringBootTest

LEVEL 3

Basic awareness

- ☐ Testcontainers
- ☐ Vitest & React Testing Library